CSA 1517

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Cloud Computing

Continuous Internal Assessment - 1

Solutions

1A Cloud Computing Adoption – Shop-Easy

(1) Cloud Computing - It provides server, storage, software and networking through the internet

(2) Evolution of Cloud Computing

- Hardware evolved from mainframes to physical servers and virtual machines
- Modern cloud technology provides resources on demand.

(3) Cloud service Models:-

(a) IAAS (Infrastructure As a service) Ex: Azure V

(b) PAAS (Parameter as a service) Ex: Google App Engi

(c) SAAS (Software as a Service) Ex:

d

4. Xaas (Anything as a Service):

Provides IT services through the Internet

Ex: . IAAS
. PaaS
. SaaS
. DBaaS
. STaaS

6) Final

| Requirement | Service Model |
|---|---|
| Legacy ERP | IAAS |
| New web Application | PAAS |
| HR Payroll & email | SAAS |
| Additional cloud service | XAAS |

2.) Given parameters

12 servers, 20% utilization

400W power for old server

500 W power per new host

70% of utilization

(a) host required

$$\frac{12 \times 0.2}{0.7} = 3.43 \cong 4$$

$4+1 = 5$ servers

(b) consolidation Ration

$$12/5 = 2.4:1$$

(c) utilisation Per Host

$$\frac{2.4}{5} \times 100 = 48\%$$

After one host failure

$$\frac{2.4}{4} \times 100 = 60\%$$

(b) Annual Electricity cost saving

Old server.

$$12 \times 400 = 4800\ W$$

New server

$$5 \times 500 = 2500\ W$$

Power saving

$$4.8 - 2.5 = 2.3\ kW$$

Annual energy saving

$$= 2.3 \times 8760 = 20148\ kWh$$

Annual cost saving

$$= 20148 \times 8$$

$$= 21,61,184$$

(3)(a)(a) CPU & Memory

Given 16 physical cores, 64 GB RAM

4 vCPU & 8 GB RAM

$$\text{CPU ration} = \frac{10 \times 4}{12} = \frac{40}{12} = 2.5:1$$

$$\text{Memory ration} = \frac{80}{60} = 1.33:1$$

Unit IV

Average Memory $10 \times 5 = 50$ GB

Sufficient 60 GB

$10 \times 4 \times 30\% = 12$

$10 \times 4 \times 50 = 20$

(b) (i) Memory

$(4 \times 2) + (6+5) = 62$

Available = 60 GB

shortage = 2 GB

(ii) CPU

$10 \times 4 \times 50\% = 20$

Available = 16

Shortage = 4

(iii) Slow performance Memory shortage

security risks

4. Cloud Federation, SOA & Privacy Issues

(1) Cloud Federation:

Cloud Federation connects two or more independent cloud systems to share resources and services.

(2) Role of SOA:-

SOA (Service-Oriented Architecture) allows different application to communicate

(3) SDDC: Software defined Data Centre manage computing storage and networking through software.

Privacy issue

(1) Unauthorized access

(2) Data leakage

(3) Identity theft

(4) Data misuse.

Security measures:-

(1) Encryption

(2) Authentication

(3) Authorization

(4) Role-based access

(5) Audit logs.

Cloud storage & Servers

(a) Daily data

200 × 720 × 5

= 72000 MB

Average bandwidth

66.67 MBPS

(b) 30-day storage cost

720 × 30 × 2

= ₹43200

(c) After 20% edge filtering

720 × 20% = 144 GB/day

Monthly storage cost:

144 × 30 × 2

= ₹8,640

Saving:

43200 − 8640 = ₹34560

(d) Serverless cost:

144000 × 20% × 30 × 0.0002

= ₹172.80

Total monthly spend

8640 + 172.80 = 8812